

PASSWORD SECURITY & AUTHENTICATION FUNDAMENTALS REPORT

Prepared By: Subodh Kumar

Date: June 2026

1. INTRODUCTION

Password security is one of the most important aspects of cybersecurity. Weak passwords can expose user accounts to unauthorized access, while strong authentication methods help protect sensitive information. This report summarizes the concepts studied during this task.

1. OBJECTIVE

The objectives of this task were:

- To understand password security fundamentals.
- To learn how weak passwords increase security risks.
- To study common authentication mechanisms.
- To explore methods used to protect user accounts.

1. PASSWORD SECURITY

Passwords serve as the first line of defense against unauthorized access. Strong passwords should contain a mix of uppercase letters, lowercase letters, numbers, and special characters.

Examples of weak passwords:

- 12345678
- password
- admin123

Examples of stronger passwords:

- Subho@#32
- G7!kP#92xL

1. PASSWORD STRENGTH ANALYSIS

A password strength test was performed to compare different password combinations.

Observations:

- Simple numeric passwords were categorized as very weak.
- Passwords with only letters and numbers provided moderate protection.
- Passwords containing uppercase letters, lowercase letters, numbers, and special characters were rated stronger.

This demonstrated the importance of password complexity and uniqueness.

1. AUTHENTICATION MECHANISMS

The following authentication methods were studied:

a) Password-Based Authentication

The user enters a username and password to access an account.

b) Two-Step Verification

A second verification step is required after entering the password.

c) One-Time Passwords (OTP)

Temporary codes are generated and used for authentication.

d) Multi-Factor Authentication (MFA)

Authentication is performed using multiple verification factors.

1. SECURITY OBSERVATIONS

- Password creation screens encourage secure password selection.
- Two-Step Verification provides an additional layer of protection.
- Security checkup tools help users monitor account activity.

- Authenticator applications strengthen account security through OTP generation.
-

1. CHALLENGES FACED

During this task, one of the main challenges was understanding how different authentication methods work together to secure user accounts. I also learned that password length alone does not guarantee security, and that factors such as complexity and multi-factor authentication play an important role in account protection.

1. CONCLUSION

This task improved my understanding of password security, authentication methods, and account protection practices. Strong passwords, OTP-based verification, and multi-factor authentication are essential for reducing the risk of unauthorized access and improving overall account security.

1. SCREENSHOT REFERENCES

Figure 1 – Password Creation Screen

Figure 2 – Two-Step Verification

Figure 3 – Google Security Checkup

Figure 4 – Authenticator Application

Figure 5 – Password Strength Analysis